

Uber

AI-Enabled Uber India Operations Analytics Dashboard Using Power BI

A Cloud-Based Business Intelligence Platform with AI-Powered Decision Support

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MODULE

Business
Intelligence

SUPERVISOR

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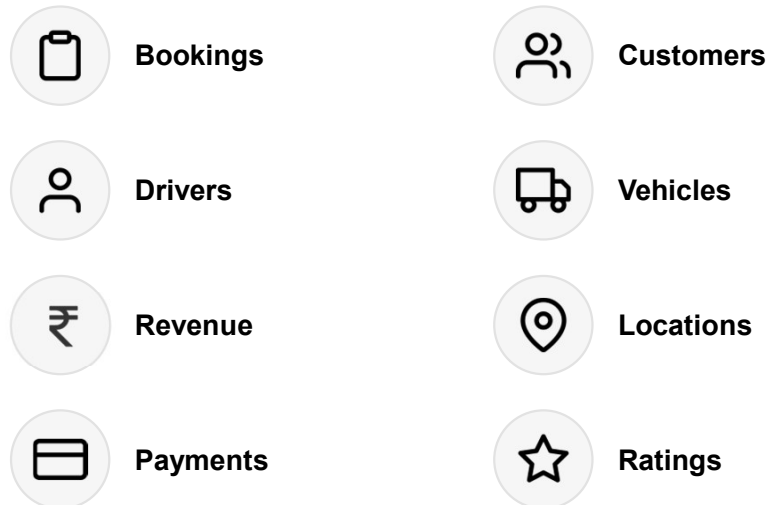
Power BI • Google Gemini 2.5 Flash • Flask • GitHub • Render Cloud

PROJECT BACKGROUND

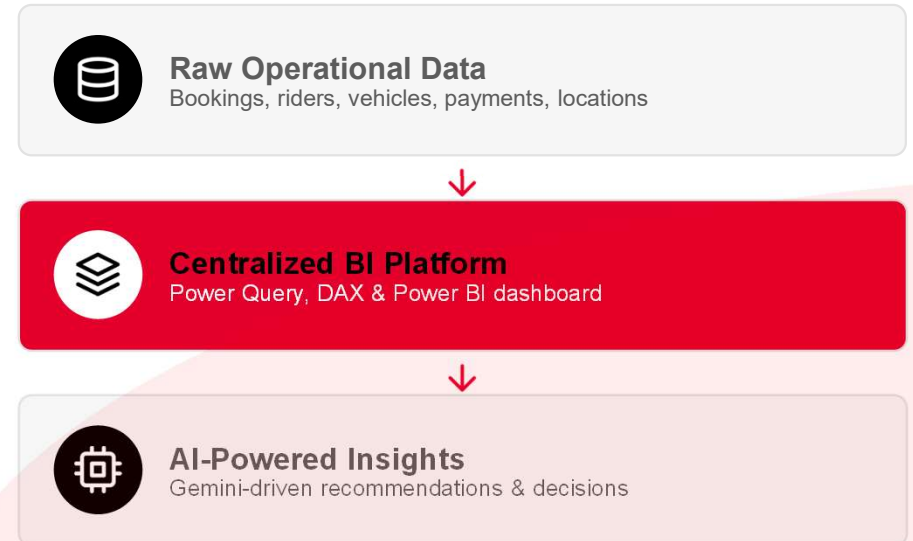
Operational Context

Every ride on the Uber platform generates a footprint of operational data — this project draws on 150,000 booking records across 12 months and 7 vehicle categories to turn that volume into timely, trustworthy insight.

Daily Operational Data Sources



From Raw Data to Decisions



Fragmented, spreadsheet-based reporting cannot keep pace with daily ride volume — Business Intelligence unifies every data source into one decision-ready view.

CURRENT STATE CHALLENGES

Business Problem



Data scattered across
Excel sheets



Difficult to identify
trends



Manual reporting
process



No centralized
dashboard



Slow decision
making



Revenue leakage
goes unnoticed



High ride
cancellations



Hard to read customer
behaviour

“Business needed a centralized AI-enabled decision support dashboard.”

SCOPE OF THE DASHBOARD

Project Objectives

Eight core objectives shaped the design of the Power BI dashboard and its AI assistant.

**Monitor Operational KPIs**

Bookings, revenue, distance at a glance

**Analyze Revenue**

By vehicle, customer & payment method

**Booking Performance**

Completed, cancelled & incomplete rides

**Vehicle Performance**

Compare Auto, Bike, Sedan, XL & more

**Customer Behaviour**

First-time, return & regular riders

**Cancellation Analysis**

Root causes behind lost bookings

**Location Analysis**

Busy pickup, drop zones & time slots

**AI-Powered Recommendations**

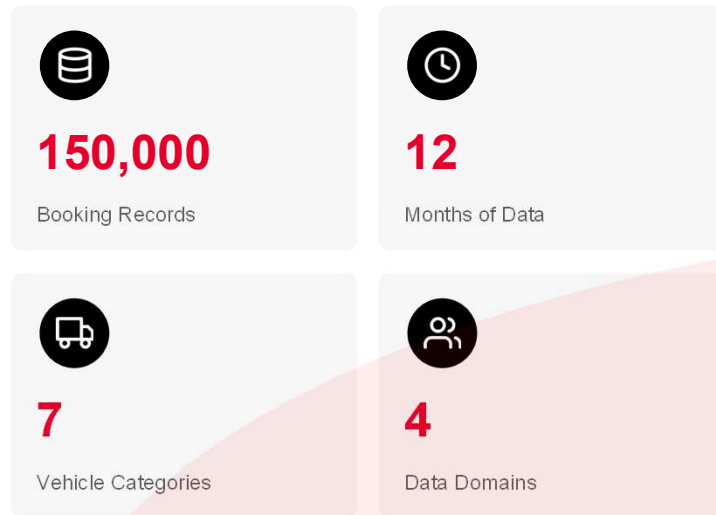
Natural-language business Q&A

DATA PIPELINE

Dataset & Methodology

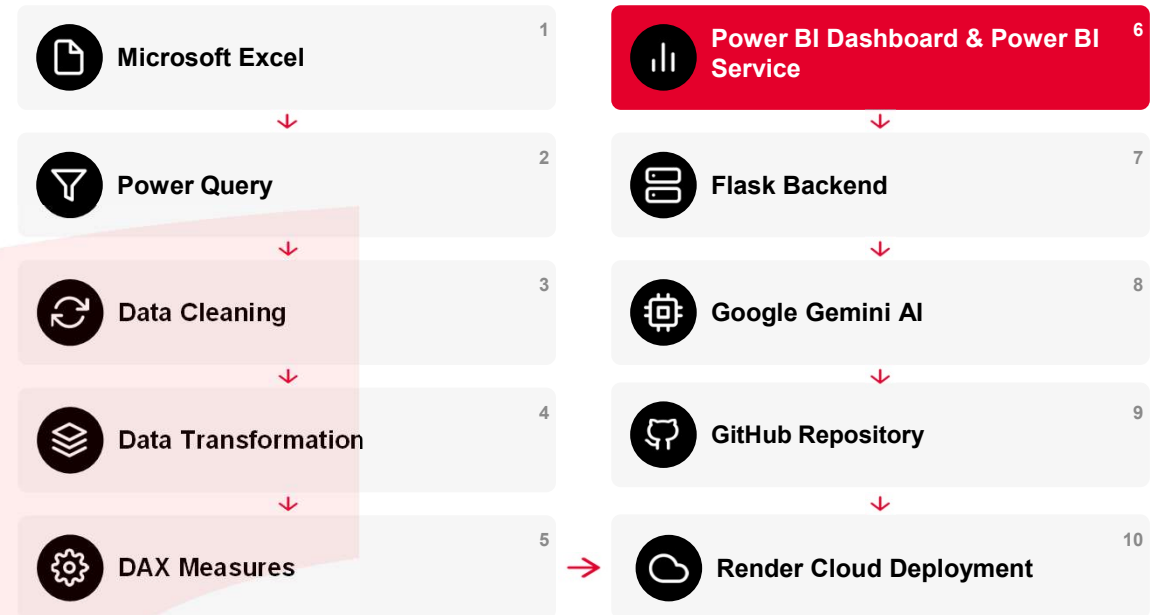
A ten-stage pipeline turns raw Excel records into a live, AI-queryable executive dashboard.

Dataset Snapshot



Customer • Revenue • Payments • Ratings

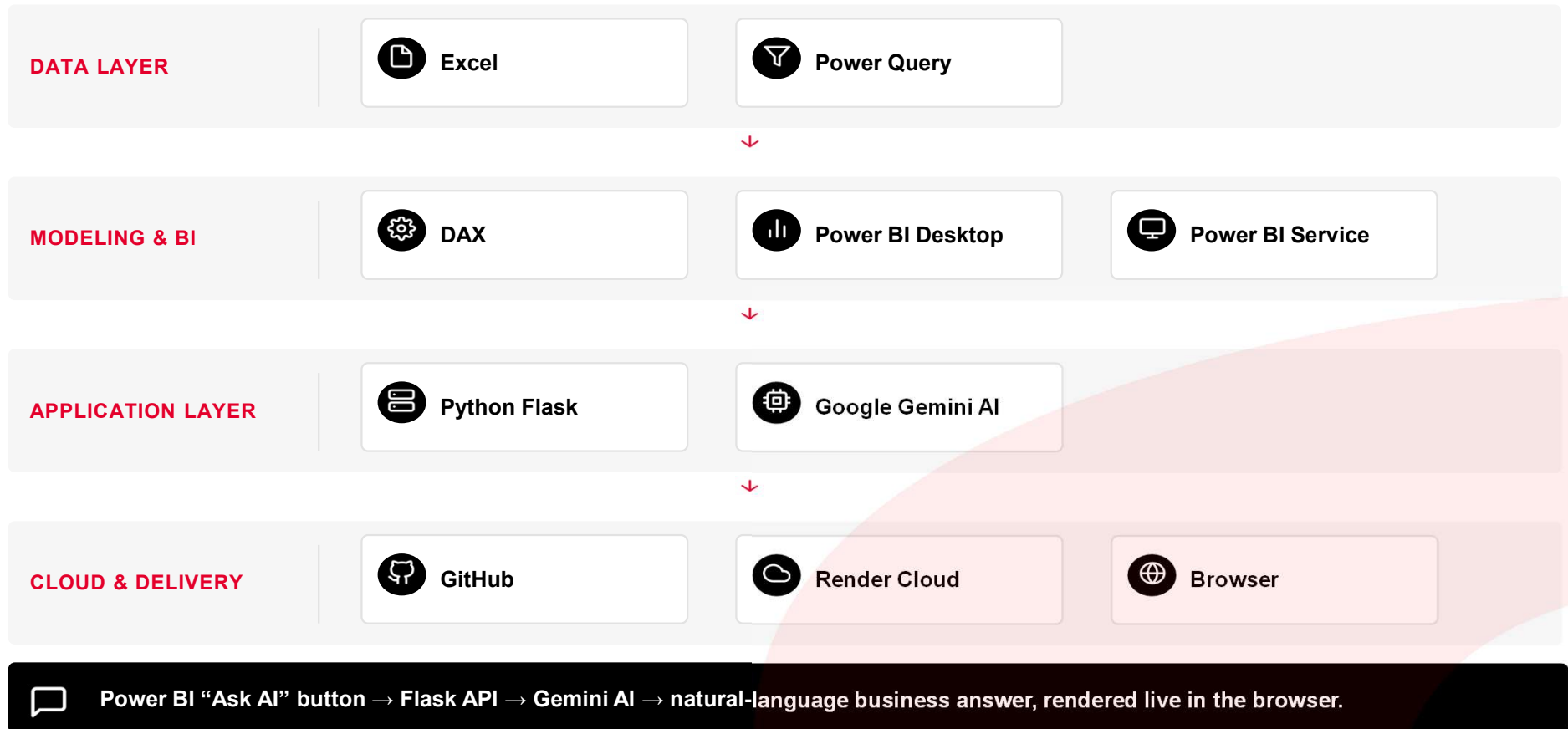
End-to-End Pipeline



SYSTEM DESIGN

Technology Stack & System Architecture

Four layers connect the raw dataset to a live, AI-powered response inside the browser.



Dashboard Overview

The dashboard is organized into five analytical modules, each built around the same five executive KPIs.

Five Analytical Modules

**Overview**
Executive summary across all operations

**Vehicle**
Performance by vehicle category

**Revenue**
Revenue by customer, vehicle & payment

**Rider**
Customer segments & cancellation reasons

**Location**
Demand hotspots & busy time slots

Executive KPIs on Every Page

**Completed Bookings**
Rides fulfilled successfully

**Lost Bookings**
Cancelled or incomplete rides

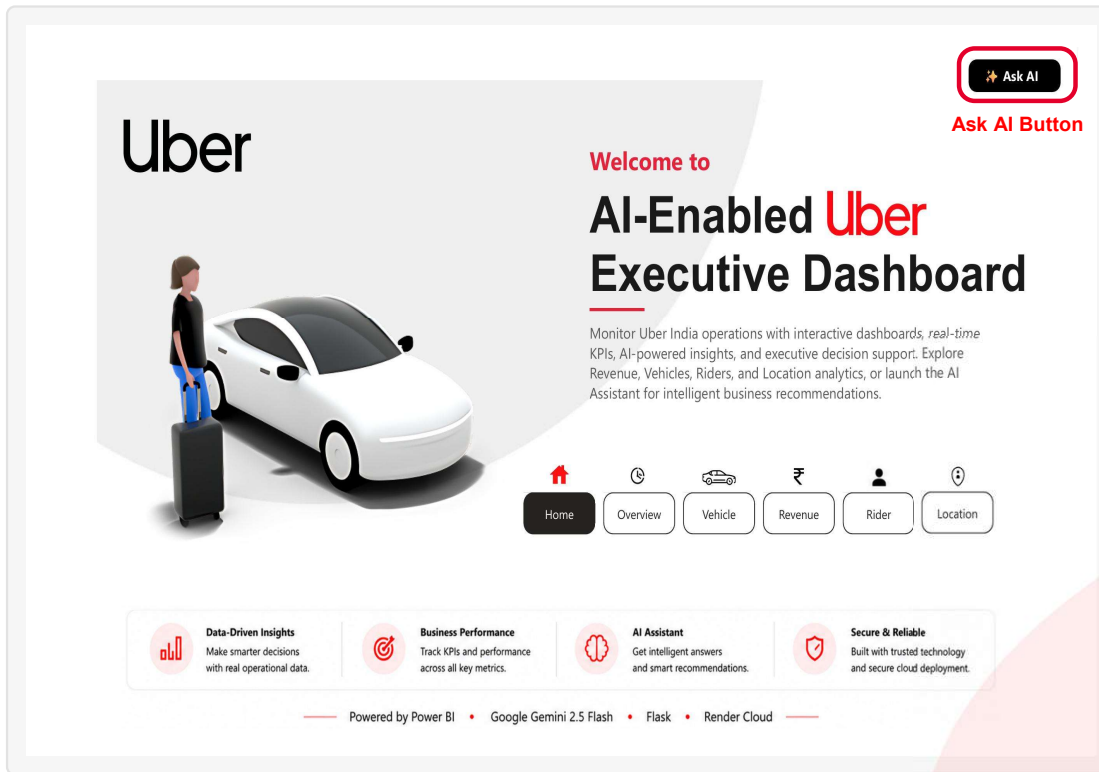
**Revenue**
Total earnings generated

**Total Distance**
Cumulative kilometres driven

**Average Distance**
Mean distance per ride

Executives don't need every raw number — they need these five metrics, consistently placed, so performance can be scanned at a glance across every module.

Dashboard Navigation & Home Page



Live Home Page — AI-Enabled Executive Dashboard

What This Page Delivers



Navigation Buttons

One-click access to Overview, Vehicle, Revenue, Rider & Location



Professional Landing Page

Uber-branded welcome screen framing the platform's purpose



Ask AI Button

Launches the live AI Assistant from anywhere in the report



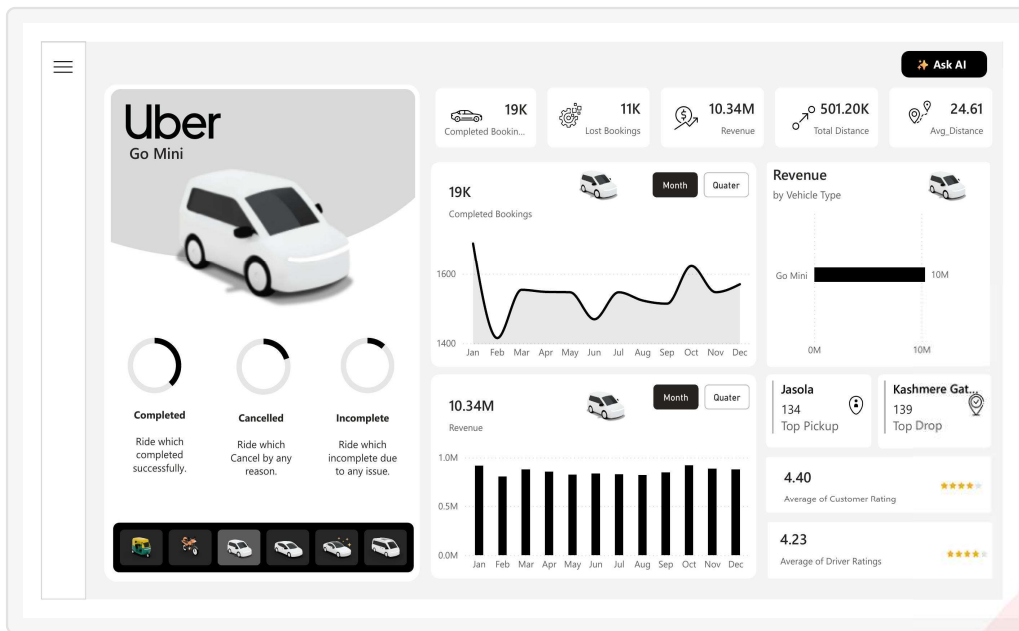
Technology Stack Footer

Power BI, Gemini 2.5 Flash, Flask & Render Cloud, credited on-page

MODULE 1 — EXECUTIVE SUMMARY

Executive Dashboard Overview

The Overview page condenses bookings, revenue, demand hotspots and service quality onto a single first screen.



Live Overview Page — Go Mini vehicle view

The Business Story



One Screen, Full Operational Health

Booking trends, revenue trends, KPI cards and ratings sit together — leadership reads business performance without opening a separate report.



Demand Concentration Is Immediately Visible

Top pickup and drop locations surface exactly where rider demand concentrates, informing where supply should be prioritized first.



Service Quality Sits Beside Volume

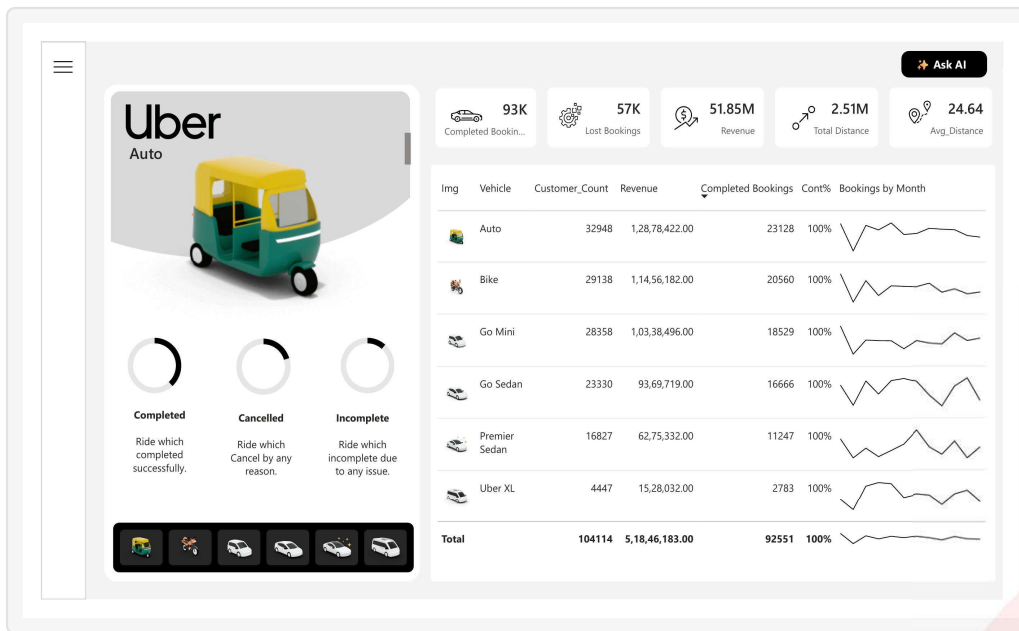
Rider and driver ratings are placed next to revenue, keeping growth decisions anchored to service quality — not volume alone.

“Executives can evaluate the overall operational health within seconds.”

MODULE 2 — FLEET PERFORMANCE

Vehicle Performance Analysis

The Vehicle page ranks every category side by side, turning fleet mix into a resourcing decision.



Live Vehicle Page — category comparison table

The Business Story



Revenue Concentration by Category

Auto and Bike together generate the largest share of completed bookings and revenue — showing where the fleet's core strength currently sits.



Contribution, Not Just Volume

The Cont% column shows each category's true share of total revenue, not just raw booking counts — separating high-volume from high-value categories.



A Basis for Resource Allocation

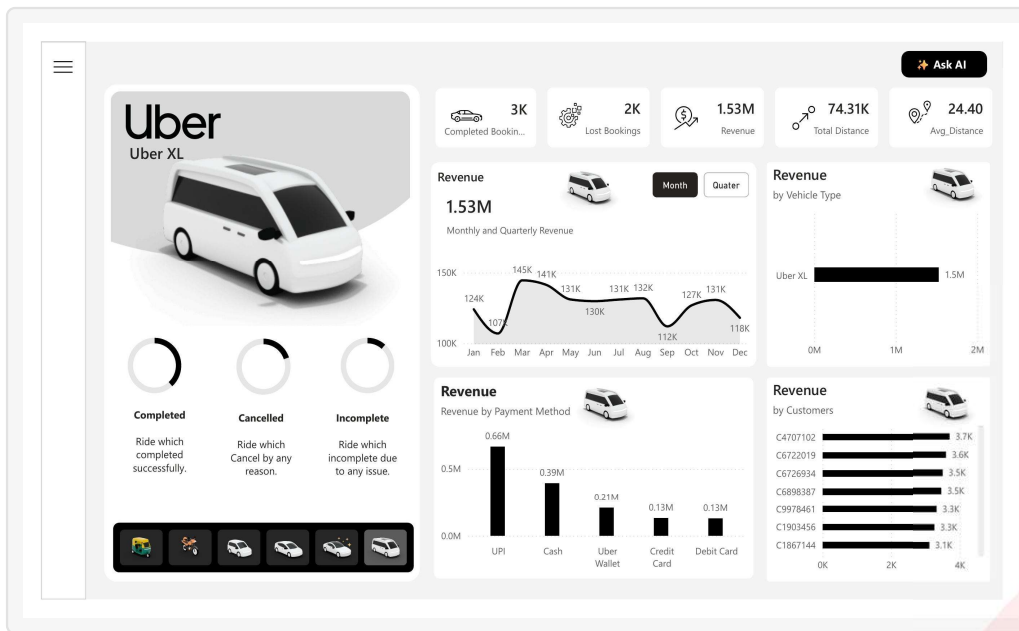
Comparing customer count against revenue per category tells management where to add supply and where demand is already well served.

“Vehicle-level analysis supports fleet optimization and operational planning.”

MODULE 3 — FINANCIAL INTELLIGENCE

Revenue Analytics

The Revenue page reframes raw transactions into payment behaviour, seasonality and customer value.



Live Revenue Page — Uber XL vehicle view

The Business Story



Payment Behaviour Reveals Channel Priorities

UPI dominates transaction volume — showing finance and product teams exactly where reliability and partnership investment matters most.



Revenue Seasonality Becomes Visible

Monthly and quarterly revenue trends expose peaks and dips finance can plan capacity and cash flow around, rather than react to after the fact.



Identifying the Premium Customer Segment

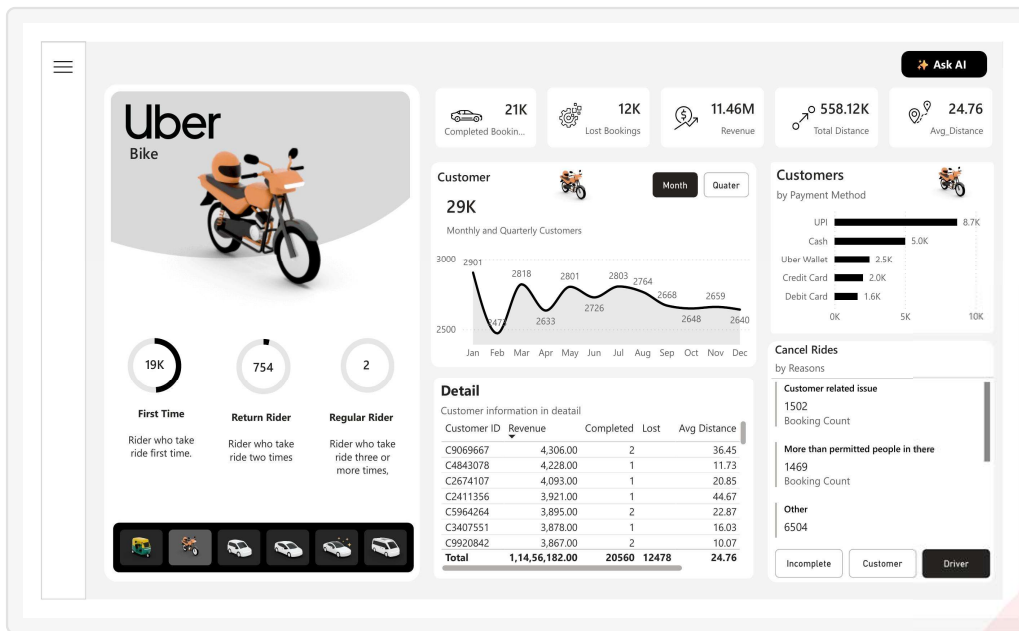
Ranking customers by revenue contribution flags a high-value segment worth dedicated retention offers and account-level engagement.

“This page transforms raw financial data into actionable business intelligence.”

MODULE 4 — CUSTOMER INTELLIGENCE

Rider Behaviour Analysis

The Rider page turns anonymous ride counts into a loyalty, retention and cancellation story.



Live Rider Page — Bike vehicle view

The Business Story



Loyalty Becomes Measurable

Segmenting first-time, return and regular riders quantifies exactly how many customers are converting into habitual, repeat users.



Cancellations Have Root Causes

Breaking cancellations down by reason turns a vague loss number into specific, fixable operational issues — not just a percentage to worry about.



Payment Preference Shapes Marketing

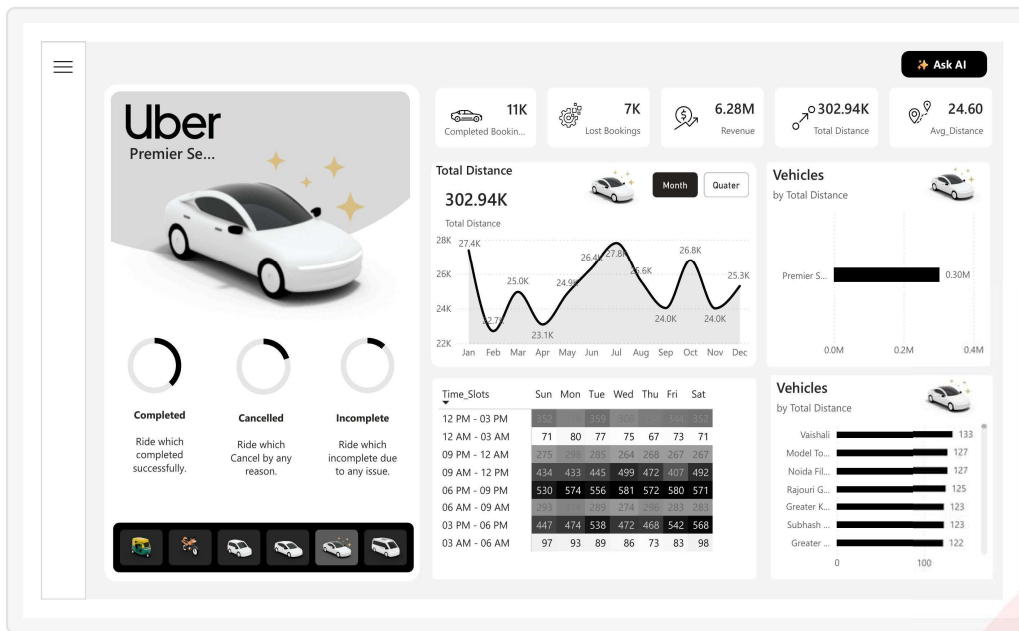
Knowing how riders actually pay determines which promotions, wallets or partnerships will genuinely reach the target audience.

“Customer analytics enables better retention and service improvement.”

MODULE 5 — OPERATIONAL INTELLIGENCE

Location Intelligence

The Location page turns geography and time into a driver-allocation and demand-forecasting tool.



Live Location Page — Premier Sedan vehicle view

The Business Story



Hotspots Show Where Supply Should Concentrate

Ranking locations by total distance highlights the specific zones generating the most driving activity across the network.



Peak Hours Drive Demand Forecasting

The time-slot heatmap exposes exactly when demand spikes across the week, so driver allocation can shift ahead of it rather than behind it.



Distance Patterns Guide Fleet Planning

Comparing distance travelled by vehicle category ties travel demand directly back to which vehicles need to be positioned, and where.

“Location intelligence enables data-driven operational planning.”

BEYOND STATIC REPORTING

AI-Powered Business Intelligence

A dashboard answers the questions it was built for. An executive's next question is rarely one of them.

TRADITIONAL DASHBOARDS



Answers Only Predefined Questions

Charts and KPIs are fixed at design time — anything outside that scope needs a brand-new report.



New Questions Mean New Development

A fresh executive question often means waiting on an analyst to build another visual from scratch.



Static By Design

Dashboards summarize what already happened; they don't reason about why, or what to do next.

THE AI ASSISTANT



Ask in Plain Language

Executives type a question the way they'd ask a colleague — no query language, no report request.



Reasons Over Business Data

The assistant sits directly on top of the same Power BI analytics platform, using the same trusted numbers.



Answers Instantly, On Demand

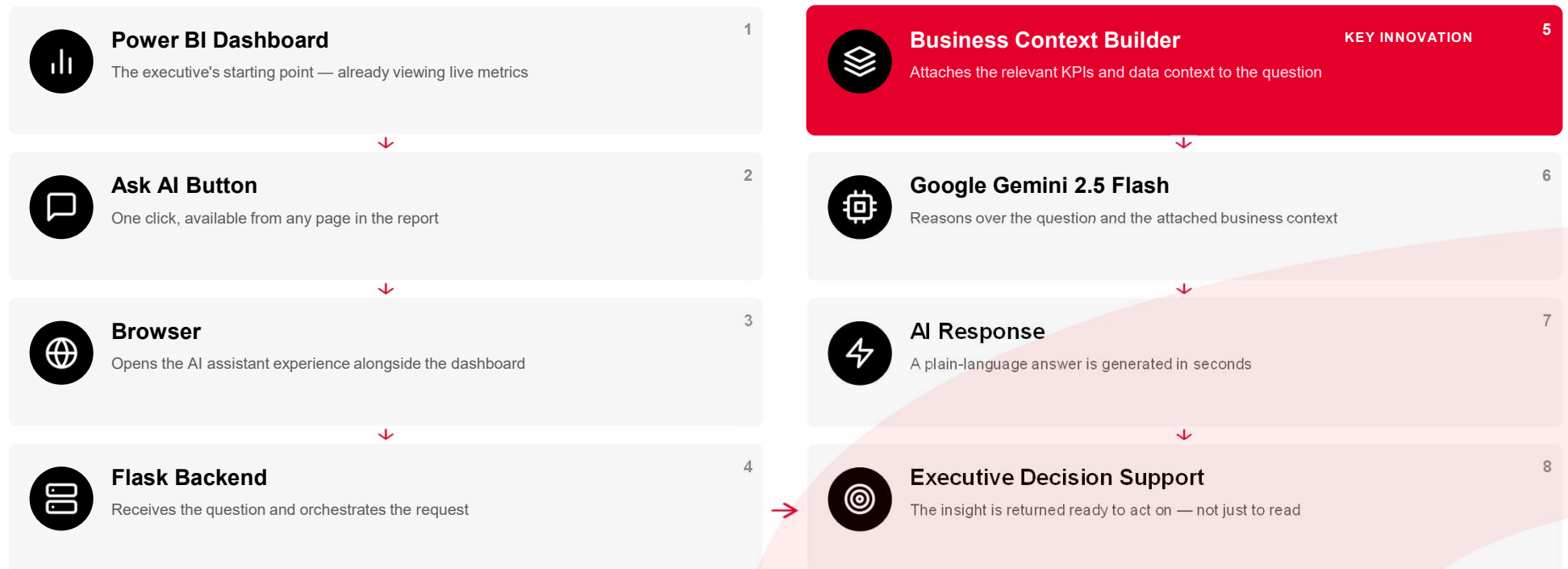
New, unplanned questions get answered in seconds — not after a follow-up report cycle.

“Business Intelligence becomes conversational.”

HOW THE CONVERSATION WORKS

AI System Architecture

Eight stages carry an executive's question from a single click to a grounded, actionable answer.

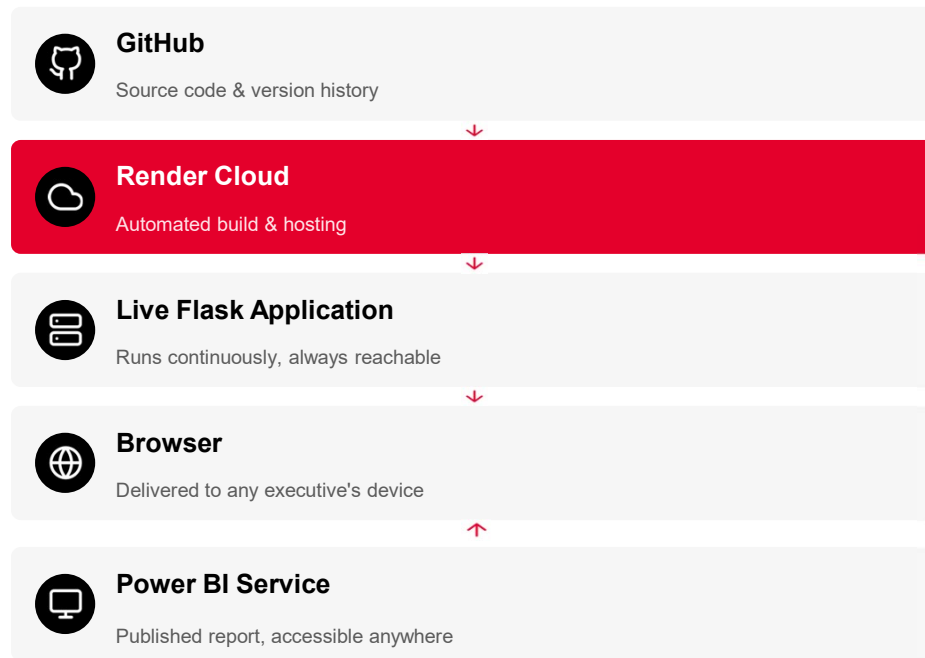


FROM DESKTOP FILE TO LIVE PLATFORM

Cloud Deployment & Live Solution

The dashboard and its AI assistant don't live on one laptop — they run continuously in the cloud.

Deployment Flow



Why This Matters



Always-On Availability

The dashboard and AI assistant run continuously, so executives reach the dashboard anytime, from any device — no desktop file to open.



Continuous Deployment

Every update pushed to the codebase deploys automatically, so improvements reach users without manual reinstallation.



Built to Scale

Cloud hosting lets the platform serve more users and requests as adoption grows, without re-architecting the system.

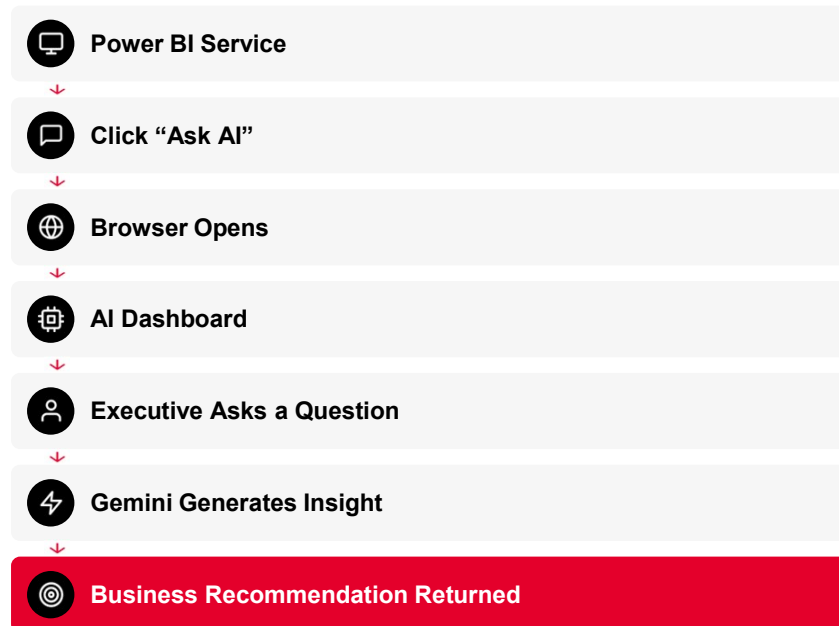
“This project evolved from a desktop dashboard into a cloud-based AI-enabled analytics platform.”

SEE IT IN ACTION

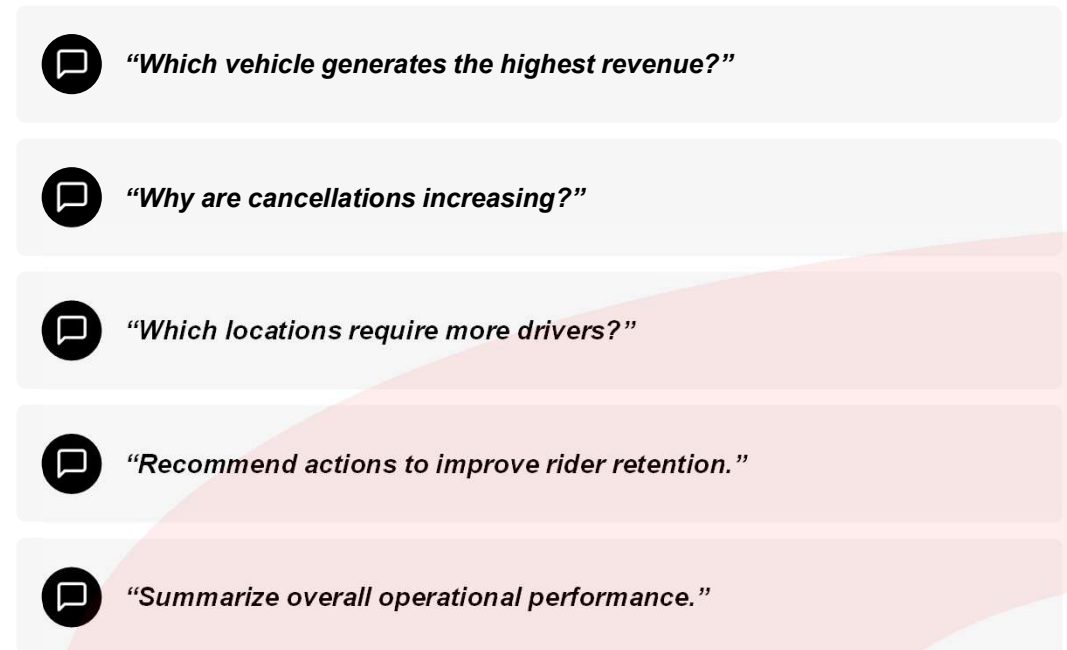
Live AI Demonstration

From a single click to a grounded recommendation — here is the complete executive user journey.

The User Journey



Example Executive Questions



"The AI Assistant transforms business analytics into executive decision support."

FROM DATA TO DECISIONS

Key Business Insights

Six executive-level observations drawn from the dashboard — beyond what any single chart shows.



Revenue Concentration

A small set of vehicle categories — Sedan and Premier — generate a disproportionate share of total revenue.



UPI Dominance

UPI is the clear preferred payment method, signalling where checkout experience investment matters most.



Customer Loyalty Patterns

A core base of regular riders drives repeat revenue, distinct from a much larger first-time rider segment.



Peak-Hour Demand Imbalance

Bookings cluster sharply around morning and evening peaks, straining driver supply at predictable times.



Location Hotspots

A handful of pickup and drop zones account for the majority of ride volume across the city.



Cancellation Trends

Cancellations concentrate around specific vehicle types and time windows, pointing to addressable root causes.

“Data becomes valuable only when transformed into actionable business insight.”

FROM INSIGHT TO ACTION

Business Recommendations

Strategic actions the dashboard and its AI assistant make possible for Uber India leadership.

01



Optimize Fleet Allocation

Shift vehicle supply toward high-revenue categories and hotspot zones identified on the Vehicle module.

02



Increase Drivers at Peak

Schedule additional drivers around the identified peak-hour windows to close the demand gap.

03



Improve Customer Retention

Target first-time riders with incentives that convert them into the regular-rider segment.

04



Reduce Cancellations

Address the specific vehicle types and time windows driving the majority of lost bookings.

05



Target High-Value Customers

Prioritize retention and service quality for the customer segment generating the most revenue.

06



Use AI for Decision-Making

Embed the AI Assistant into executive workflows for on-demand, natural-language business answers.

“Analytics should drive decisions—not just reports.”

LOOKING BACK, LOOKING FORWARD

Challenges, Future Scope & Conclusion

What the build tested along the way, where the platform goes next, and what it proves today.

IMPLEMENTATION CHALLENGES

-  Learning Power BI & BI Concepts
-  AI Integration (Gemini + Flask)
-  Learning Python & Cloud Tools
-  Cloud Deployment
-  System Testing & Validation

FUTURE SCOPE

-  Multi-country Support
-  Excel/CSV Upload
-  Advanced AI Recommendations
-  Role-based Authentication
-  Industry Expansion

This project demonstrates that fragmented, spreadsheet-based reporting can be transformed into a live, cloud-based, AI-enabled Business Intelligence platform — one capable of genuine executive decision-making, and proof that the objectives set out on the opening slide have been fully achieved.

THANK YOU

Questions & Discussion

AI-Enabled Uber India Operations Analytics Dashboard

Power BI | Google Gemini AI | Flask | GitHub | Render

